GAUSSIAN INTEGRALS AND GAUSSIAN PROCESSES PROBLEM SHEET

This sheet goes with the lectures on Gaussian integrals and Gaussian processes. Questions 1 and 2 only ask you to *use* the results below; Questions 3 and 4 derive them, and are longer and harder. Do Questions 1 and 2 first.

The Gaussian toolkit

- **Integrals:** $\int_{-\infty}^{\infty} \mathrm{e}^{-\frac{a}{2}x^2+bx}\,\mathrm{d}x = \sqrt{\frac{2\pi}{a}}\,\mathrm{e}^{\frac{b^2}{2a}}$ for $a>0$, and in n dimensions, for positive definite $\mathbf{A}$, $\int \mathrm{e}^{-\frac{1}{2}\boldsymbol{x}^\mathsf{T}\mathbf{A}\boldsymbol{x}+\boldsymbol{b}^\mathsf{T}\boldsymbol{x}}\,\mathrm{d}\boldsymbol{x} = \sqrt{\frac{(2\pi)^n}{\det(\mathbf{A})}}\,\mathrm{e}^{\frac{1}{2}\boldsymbol{b}^\mathsf{T}\mathbf{A}^{-1}\boldsymbol{b}}$.
- **Moment generating function:** if $\boldsymbol{x}\sim\mathcal{N}(\boldsymbol{\mu},\boldsymbol{\Sigma})$ then $\mathbb{E}\left[\mathrm{e}^{\boldsymbol{t}^\mathsf{T}\boldsymbol{x}}\right] = \mathrm{e}^{\boldsymbol{t}^\mathsf{T}\boldsymbol{\mu}+\frac{1}{2}\boldsymbol{t}^\mathsf{T}\boldsymbol{\Sigma}\boldsymbol{t}}$ (in one dimension $\mathbb{E}\left[\mathrm{e}^{tX}\right] = \mathrm{e}^{\mu t+\frac{1}{2}\sigma^2t^2}$).
- **Sums:** if $X\sim\mathcal{N}(\mu_1,\sigma_1^2)$ and $Y\sim\mathcal{N}(\mu_2,\sigma_2^2)$ are independent then $X+Y\sim\mathcal{N}(\mu_1+\mu_2,\sigma_1^2+\sigma_2^2)$.
- **Linear maps:** $\mathbf{W}\boldsymbol{x}+\boldsymbol{c}\sim\mathcal{N}(\mathbf{W}\boldsymbol{\mu}+\boldsymbol{c},\ \mathbf{W}\boldsymbol{\Sigma}\mathbf{W}^\mathsf{T})$.
- **Marginals and conditionals:** for $\begin{pmatrix}\boldsymbol{z}_1\\ \boldsymbol{z}_2\end{pmatrix}\sim\mathcal{N}\left(\begin{pmatrix}\boldsymbol{\mu}_1\\ \boldsymbol{\mu}_2\end{pmatrix},\begin{pmatrix}\boldsymbol{\Sigma}_{11} & \boldsymbol{\Sigma}_{12}\\ \boldsymbol{\Sigma}_{21} & \boldsymbol{\Sigma}_{22}\end{pmatrix}\right)$:
  - marginal: $\boldsymbol{z}_1\sim\mathcal{N}(\boldsymbol{\mu}_1,\boldsymbol{\Sigma}_{11})$ (strike out the rows and columns of $\boldsymbol{z}_2$);
  - conditional: $\boldsymbol{z}_1\mid\boldsymbol{z}_2\sim\mathcal{N}\big(\boldsymbol{\mu}_1+\boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}(\boldsymbol{z}_2-\boldsymbol{\mu}_2),\ \boldsymbol{\Sigma}_{11}-\boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}\boldsymbol{\Sigma}_{21}\big)$.

1 Each part needs only one or two of the results in the box.

(a) Evaluate $\int_{-\infty}^{\infty}\mathrm{e}^{-x^2/2+3x}\,\mathrm{d}x$.

Result 1 with $a=1, b=3$: $\sqrt{2\pi}\,\mathrm{e}^{9/2}\approx 225.6$.

[2 marks]

(b) $X\sim\mathcal{N}(1,4)$. Use the moment generating function to find $\mathbb{E}\left[\mathrm{e}^X\right]$, and $\mathbb{E}\left[X^2\right]$.

$\mathbb{E}\left[\mathrm{e}^{tX}\right] = \mathrm{e}^{t+2t^2}$, so at $t=1$, $\mathbb{E}\left[\mathrm{e}^X\right] = \mathrm{e}^3$. Differentiating twice at $t=0$, or using $\mathbb{E}\left[X^2\right] = \mu^2+\sigma^2$, gives $\mathbb{E}\left[X^2\right] = 1+4 = 5$. (Note $\mathbb{E}\left[\mathrm{e}^X\right] \neq \mathrm{e}^{\mathbb{E}\left[X\right]} = \mathrm{e}^1$.)

[3 marks]

Answers

(c) $X \sim \mathcal{N}(1,1)$ and $Y \sim \mathcal{N}(2,3)$ are independent. What are the distributions of $X+Y$ and of $2X-Y$?

$X+Y \sim \mathcal{N}(3,4)$. For $2X-Y$, the mean is $2\times 1-2=0$ and the variance is $2^2\times 1+(-1)^2\times 3=7$, so $2X-Y\sim\mathcal{N}(0,7)$. (Variances add even when we subtract.)

[3 marks]

(d) $\boldsymbol{x}\sim\mathcal{N}(\mathbf{0},\mathbf{I})$ in two dimensions and $\boldsymbol{z}=\mathbf{W}\boldsymbol{x}$ with $\mathbf{W}=\begin{pmatrix}1 & 1\\ 1 & -1\end{pmatrix}$. What is the distribution of $\boldsymbol{z}$? Are $z_1=x_1+x_2$ and $z_2=x_1-x_2$ independent?

$\boldsymbol{z}\sim\mathcal{N}(\mathbf{0},\mathbf{W}\mathbf{W}^\mathsf{T})$ with $\mathbf{W}\mathbf{W}^\mathsf{T}=\begin{pmatrix}2 & 0\\ 0 & 2\end{pmatrix}$. The covariance is diagonal, and for jointly Gaussian variables uncorrelated means independent, so yes.

[3 marks]

(e) $\boldsymbol{z}\sim\mathcal{N}(\boldsymbol{\mu},\boldsymbol{\Sigma})$ with $\boldsymbol{\mu}=(1,2,3)^\mathsf{T}$ and $\boldsymbol{\Sigma}=\begin{pmatrix}4 & 1 & 0\\ 1 & 3 & 1\\ 0 & 1 & 2\end{pmatrix}$. What is the marginal distribution of (z_1,z_3)?

Strike out the second row and column: $(z_1,z_3)\sim\mathcal{N}\left((1,3)^\mathsf{T},\begin{pmatrix}4 & 0\\ 0 & 2\end{pmatrix}\right)$, so z_1 and z_3 are independent, even though both are correlated with z_2.

[2 marks]

(f) (z_1,z_2) is Gaussian with mean $\mathbf{0}$ and covariance $\begin{pmatrix}2 & 1\\ 1 & 1\end{pmatrix}$. We observe $z_2=2$. What is the distribution of z_1 now? Compare it with the distribution of z_1 before the observation.

Mean $0+1\times 1^{-1}\times(2-0)=2$; variance $2-1\times 1^{-1}\times 1=1$. So $z_1\mid z_2=2\sim\mathcal{N}(2,1)$: the observation moved the mean from 0 to 2 and halved the variance (from 2).

[3 marks]

(g) Our prior belief about a quantity is $x\sim\mathcal{N}(0,4)$. We measure $y=x+\epsilon$ with noise $\epsilon\sim\mathcal{N}(0,1)$ and get $y=3$. Write down the joint distribution of (x,y) and hence find the distribution of x given $y=3$.

y has mean 0 and variance $4+1=5$, and $\mathrm{Cov}(x,y)=\mathrm{Var}(x)=4$, so $(x,y)\sim\mathcal{N}\left(\mathbf{0},\begin{pmatrix}4 & 4\\ 4 & 5\end{pmatrix}\right)$. Conditioning, $\mathbb{E}[x\mid y]=\frac{4}{5}\times 3=2.4$ and $\mathrm{Var}(x\mid y)=4-\frac{16}{5}=0.8$. The estimate is pulled most of the way to the measurement, and $1/0.8=1/4+1/1$: precisions add.

[4 marks]

End of question 1

Answers

2 A Gaussian process with covariance function $K(x,x') = \mathrm{e}^{-(x-x')^2/2}$ has, for observations $\boldsymbol{y}$ at inputs $x_1,\ldots,x_m$ with noise variance σ^2, the predictive distribution at a new input x_*

$$f_* \mid \boldsymbol{y} \sim \mathcal{N}\big(\boldsymbol{k}_*^\mathsf{T}(\mathbf{K}+\sigma^2\mathbf{I})^{-1}\boldsymbol{y},\; k_* - \boldsymbol{k}_*^\mathsf{T}(\mathbf{K}+\sigma^2\mathbf{I})^{-1}\boldsymbol{k}_*\big)$$

where $K_{ij} = K(x_i,x_j)$, $(\boldsymbol{k}_*)_i = K(x_i,x_*)$ and $k_* = K(x_*,x_*)$.

(a) Explain why this is the conditional result in the box, with $\boldsymbol{z}_1 = f_*$ and $\boldsymbol{z}_2 = \boldsymbol{y}$. What are $\boldsymbol{\Sigma}_{11}$, $\boldsymbol{\Sigma}_{12}$ and $\boldsymbol{\Sigma}_{22}$?

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f_* and the observations are jointly Gaussian with mean zero. $\boldsymbol{\Sigma}_{11} = k_*$ (the prior variance of f_*), $\boldsymbol{\Sigma}_{12} = \boldsymbol{k}_*^\mathsf{T}$ (the covariance of f_* with each observation) and $\boldsymbol{\Sigma}_{22} = \mathbf{K} + \sigma^2\mathbf{I}$ (the observations' covariance, with the noise variance added on the diagonal). Substituting into the conditional formula gives the result.

[3 marks]

(b) Take a single noise-free observation ($\sigma^2 = 0$), $y_1 = 1$ at $x_1 = 0$. Find the predictive mean and variance at x_*. What are they at $x_* = 0$, and as $x_* \to \infty$?

$\mathbf{K} = 1$ and $k_* = 1$, and $\boldsymbol{k}_* = \mathrm{e}^{-x_*^2/2}$. So the mean is $\mathrm{e}^{-x_*^2/2}$ and the variance is $1 - \mathrm{e}^{-x_*^2}$. At $x_* = 0$ the mean is 1 and the variance 0: the prediction passes through the data point exactly. As $x_* \to \infty$ the mean tends to 0 and the variance to 1, the prior: far from the data we know nothing more than before.

[4 marks]

(c) Repeat with noise $\sigma^2 = 1$. What changes at $x_* = 0$?

Now the mean is $\mathrm{e}^{-x_*^2/2}/2$ and the variance $1 - \mathrm{e}^{-x_*^2}/2$. At $x_* = 0$ the mean is $\frac{1}{2}$ and the variance $\frac{1}{2}$: a noisy observation is only partly trusted (this is the noisy-measurement result of Question 1(g)).

[3 marks]

(d) Now take two noise-free observations, $y = 1$ at $x = -1$ and $y = 1$ at $x = 1$. Find the predictive mean and variance at $x_* = 0$. (Hint: $\boldsymbol{y} = (1,1)^\mathsf{T}$ is an eigenvector of $\mathbf{K}$.)

$\mathbf{K} = \begin{pmatrix} 1 & \mathrm{e}^{-2} \\ \mathrm{e}^{-2} & 1 \end{pmatrix}$ and $\boldsymbol{k}_* = \mathrm{e}^{-1/2}(1,1)^\mathsf{T}$. Since $\mathbf{K}(1,1)^\mathsf{T} = (1+\mathrm{e}^{-2})(1,1)^\mathsf{T}$, $\mathbf{K}^{-1}(1,1)^\mathsf{T} = (1,1)^\mathsf{T}/(1+\mathrm{e}^{-2})$. So the mean is $2\mathrm{e}^{-1/2}/(1+\mathrm{e}^{-2}) \approx 1.068$ and the variance is $1 - 2\mathrm{e}^{-1}/(1+\mathrm{e}^{-2}) \approx 0.352$. Between two points the mean overshoots 1 slightly, and the uncertainty is smaller than with one point but not zero.

[5 marks]

End of question 2

3 This question derives the normalisation of the multivariate normal step by step.

(a) Consider the integral

$$I_1 = \int_{-\infty}^{\infty} \mathrm{e}^{-x^2/2} \mathrm{d}x.$$

Directly evaluating this is difficult, but there is a trick. Consider instead

$$I_1^2 = \int_{-\infty}^{\infty} \mathrm{e}^{-x^2/2} \mathrm{d}x \int_{-\infty}^{\infty} \mathrm{e}^{-y^2/2} \mathrm{d}y = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \mathrm{e}^{-(x^2+y^2)/2} \mathrm{d}x \mathrm{d}y.$$

By making the change of variables to polar coordinates where $r = \sqrt{x^2 + y^2}$ and $\theta = \arctan(y/x)$ (so that $x = r\cos(\theta)$, $y = r\sin(\theta)$) then $\mathrm{d}x\mathrm{d}y = r\mathrm{d}r\mathrm{d}\theta$. Note that to integrate over all space we let θ vary from 0 to 2π and r to vary from 0 to ∞. Write down the integral in polar coordinates, make the further change of variables $u = r^2/2$ to evaluate I_1^2, and hence compute I_1.

$$I_1^2 = \int_0^{2\pi} \mathrm{d}\theta \int_0^{\infty} re^{-r^2/2} \mathrm{d}r = 2\pi \int_0^{\infty} re^{-r^2/2} \mathrm{d}r.$$

Making the change of variables $u = r^2/2$ so that $\mathrm{d}u/\mathrm{d}r = r$ or $\mathrm{d}u = r\mathrm{d}r$ we get

$$I_1^2 = 2\pi \int_0^{\infty} e^{-u} \mathrm{d}u = 2\pi$$

so $I_1 = \sqrt{2\pi}$.

[5 marks]

(b) By making a change of variables compute

$$I_2 = \int_{-\infty}^{\infty} \mathrm{e}^{-(x-\mu)^2/(2\sigma^2)} \mathrm{d}x$$

The trick is to let $z = (x - \mu)/\sigma$ so that $\mathrm{d}z = \mathrm{d}x/\sigma$ or $\mathrm{d}x = \sigma\mathrm{d}z$. Then

$$I_2 = \sigma \int_{-\infty}^{\infty} \mathrm{e}^{-z^2/2} \mathrm{d}z = \sigma I_1 = \sqrt{2\pi}\sigma$$

Note that the *probability density function* (PDF) for a normally distributed random variable is given by

$$\mathcal{N}(x \mid \mu, \sigma^2) = \frac{1}{\sqrt{2\pi}\sigma} \mathrm{e}^{-(x-\mu)^2/(2\sigma^2)}.$$

(Observe that at its peak $\mathcal{N}(\mu \mid \mu, \sigma^2) = 1/(\sqrt{2\pi}\sigma)$, so when $\sigma < 1/\sqrt{2\pi}$ the density exceeds 1, showing that PDFs are not probabilities.)

[5 marks]

Answers

(c) By using the identity $\mathrm{e}^{a+b} = \mathrm{e}^a \mathrm{e}^b$, or more generally

$$\mathrm{e}^{\sum_i a_i} = \prod_i \mathrm{e}^{a_i}$$

compute

$$I_3 = \int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} \mathrm{e}^{-\frac{1}{2}\|\boldsymbol{x}\|_2^2} \mathrm{d}x_1 \cdots \mathrm{d}x_n$$

where $\boldsymbol{x} = (x_1, x_2, \ldots, x_n)^{\mathsf{T}}$.

We note that $\|\boldsymbol{x}\|_2^2 = \sum_{i=1}^{n} x_i^2$

$$\begin{aligned} I_3 &= \int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} \mathrm{e}^{-\frac{1}{2}\sum_{i=1}^{n} x_i^2} \mathrm{d}x_1 \cdots \mathrm{d}x_n \\ &= \prod_{i=1}^{n} \int_{-\infty}^{\infty} \mathrm{e}^{-x_i^2/2} \mathrm{d}x_i = \prod_{i=1}^{n} \sqrt{2\pi} = (2\pi)^{n/2}. \end{aligned}$$

[3 marks]

Answers

(d) By using the fact that for a positive definite matrix, $\boldsymbol{\Sigma}$, we can use the eigenvector decomposition $\boldsymbol{\Sigma}^{-1} = \mathbf{V}\boldsymbol{\Lambda}^{-1}\mathbf{V}^{\mathsf{T}}$ where $\mathbf{V}$ is an orthogonal matrix with determinant $\det(\mathbf{V}) = \pm 1$ and $\boldsymbol{\Lambda}^{-1}$ is a diagonal matrix with elements λ_i^{-1} compute

$$I_4 = \int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} \mathrm{e}^{-\frac{1}{2}(\boldsymbol{x}-\boldsymbol{\mu})^{\mathsf{T}}\boldsymbol{\Sigma}^{-1}(\boldsymbol{x}-\boldsymbol{\mu})} \mathrm{d}x_1 \cdots \mathrm{d}x_n.$$

This needs some confidence to push through to the end.

$$I_4 = \int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} \mathrm{e}^{-\frac{1}{2}(\boldsymbol{x}-\boldsymbol{\mu})^{\mathsf{T}}\mathbf{V}\boldsymbol{\Lambda}^{-1}\mathbf{V}^{\mathsf{T}}(\boldsymbol{x}-\boldsymbol{\mu})} \mathrm{d}x_1 \cdots \mathrm{d}x_n.$$

We make the change of variables $\boldsymbol{y} = \mathbf{V}^{\mathsf{T}}(\boldsymbol{x} - \boldsymbol{\mu})$. However, to make a change of variables, $\boldsymbol{x} \to \boldsymbol{y}$, where we use

$$\iiint_{\mathcal{R}(\boldsymbol{x})} f(\boldsymbol{x}) \mathrm{d}x_1 \cdots \mathrm{d}x_n = \iiint_{\mathcal{R}(\boldsymbol{y})} f(\boldsymbol{x}(\boldsymbol{y})) |\mathbf{J}| \mathrm{d}y_1 \cdots \mathrm{d}y_n$$

where $\mathcal{R}(\boldsymbol{y})$ is the same region as $\mathcal{R}(\boldsymbol{x})$ but specified in $\boldsymbol{y}$-coordinates and $\mathbf{J}$ is the Jacobian matrix with elements $J_{ij} = \partial x_i / \partial y_j$. In our case $\boldsymbol{x}(\boldsymbol{y}) = \mathbf{V}\boldsymbol{y} + \boldsymbol{\mu}$

$$J_{ij} = \frac{\partial x_i}{\partial y_j} = \frac{\partial}{\partial y_j}\left(\sum_{k=1}^{n} V_{ik} y_k + \mu_i\right) = V_{ij}$$

so that $\mathbf{J} = \mathbf{V}$ and $|\det(\mathbf{J})| = |\det(\mathbf{V})| = 1$. This makes sense as the matrix $\mathbf{V}$ corresponds to a rotation (with a possible reflection) which does not change the volume. Thus on making the change of variables

$$I_4 = \int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} \mathrm{e}^{-\frac{1}{2}\boldsymbol{y}^{\mathsf{T}}\boldsymbol{\Lambda}^{-1}\boldsymbol{y}} \mathrm{d}y_1 \cdots \mathrm{d}y_n. = \prod_{i=1}^{n} \int_{-\infty}^{\infty} \mathrm{e}^{-y_i^2/(2\lambda_i)} \mathrm{d}y_i = \prod_i \sqrt{2\pi\lambda_i}$$

where we used I_2 with $\sigma = \sqrt{\lambda_i}$. (If $\boldsymbol{\Sigma}$ were not positive definite some $\lambda_i \leq 0$ and the integral would not converge.)

[6 marks]

Answers

• Do not write in this space •

(e) Using the facts, that $\mathbf{\Sigma} = \mathbf{V\Lambda V}^\mathsf{T}$, for any two square matrices $\mathbf{A}$ and $\mathbf{B}$ the determinants satisfy $\det(\mathbf{AB}) = \det(\mathbf{A})\det(\mathbf{B})$, and $\det(\mathbf{V}) = \det(\mathbf{V}^\mathsf{T}) = \pm 1$ show that $\det(\mathbf{\Sigma}) = \prod_i \lambda_i$.

$$\det(\mathbf{\Sigma}) = \det(\mathbf{V\Lambda V}^\mathsf{T}) = \det(\mathbf{V})\det(\mathbf{\Lambda})\det(\mathbf{V}^\mathsf{T}) = \det(\mathbf{\Lambda}) = \prod_{i=1}^{n} \lambda_i$$

We have used that $\det(\mathbf{V}) = \det(\mathbf{V}^\mathsf{T})$ which equals 1 or -1, but $\det(\mathbf{V}) \times \det(\mathbf{V}^\mathsf{T}) = 1$. Also we use that the determinant of a diagonal matrix is equal to the product of the diagonal elements. We note that

$$I_4 = (2\pi)^{n/2}\sqrt{\det(\mathbf{\Sigma})}$$

There is another trick to simplify the notation a bit more. For an $n \times n$ matrix $\mathbf{M}$ then $\det(c\mathbf{M}) = c^n \det(\mathbf{M})$ as the determinant of the matrix involves a sum of terms where each term involves a product of n elements. Multiplying each element by c means that the determinant increases by c^n. Thus, we can write

$$I_4 = (2\pi)^{n/2}\sqrt{\det(\mathbf{\Sigma})} = \sqrt{\det(2\pi\mathbf{\Sigma})}$$

and the multivariate normal PDF is

$$\mathcal{N}(\boldsymbol{x} \mid \boldsymbol{\mu}, \mathbf{\Sigma}) = \frac{1}{\sqrt{\det(2\pi\mathbf{\Sigma})}} \mathrm{e}^{-\frac{1}{2}(\boldsymbol{x}-\boldsymbol{\mu})^\mathsf{T}\mathbf{\Sigma}^{-1}(\boldsymbol{x}-\boldsymbol{\mu})}$$

[1 mark]

End of question 3

Answers

4 This question is harder: it derives the marginal and conditional results in the box by doing the integrals directly (the lecture used a shorter route). Take it slowly; each part is a short calculation.

Consider a multivariate normal distribution

$$f_{\boldsymbol{X},\boldsymbol{Y}}(\boldsymbol{x},\boldsymbol{y}) = \mathcal{N}\left(\begin{pmatrix}\boldsymbol{x}\\ \boldsymbol{y}\end{pmatrix}\middle|\begin{pmatrix}\mathbf{0}\\ \mathbf{0}\end{pmatrix}, \begin{pmatrix}\mathbf{A} & \mathbf{B}\\ \mathbf{B}^{\mathsf{T}} & \mathbf{C}\end{pmatrix}\right)$$

where $\mathbf{A}$ and $\mathbf{C}$ are symmetric (positive definite) matrices. The matrix

$$\boldsymbol{\Sigma} = \begin{pmatrix}\mathbf{A} & \mathbf{B}\\ \mathbf{B}^{\mathsf{T}} & \mathbf{C}\end{pmatrix}$$

is the covariance matrix.

We want to compute the conditional probability density function $f_{\boldsymbol{X},\boldsymbol{Y}}(\boldsymbol{x} \mid \boldsymbol{y})$. This is complicated because the normal distribution involves the inverse of the covariance matrix. Let

$$\mathbf{U} = \begin{pmatrix}\mathbf{I} & \mathbf{B}\mathbf{C}^{-1}\\ \mathbf{0} & \mathbf{I}\end{pmatrix}, \qquad \mathbf{D} = \begin{pmatrix}\mathbf{A}-\mathbf{B}\mathbf{C}^{-1}\mathbf{B}^{\mathsf{T}} & \mathbf{0}\\ \mathbf{0}^{\mathsf{T}} & \mathbf{C}\end{pmatrix}$$

where $\mathbf{I}$ is the identity matrix.

(a) Compute $\mathbf{UD}$

$$\mathbf{UD} = \begin{pmatrix}\mathbf{I} & \mathbf{B}\mathbf{C}^{-1}\\ \mathbf{0} & \mathbf{I}\end{pmatrix}\begin{pmatrix}\mathbf{A}-\mathbf{B}\mathbf{C}^{-1}\mathbf{B}^{\mathsf{T}} & \mathbf{0}\\ \mathbf{0}^{\mathsf{T}} & \mathbf{C}\end{pmatrix} = \begin{pmatrix}\mathbf{A}-\mathbf{B}\mathbf{C}^{-1}\mathbf{B}^{\mathsf{T}} & \mathbf{B}\\ \mathbf{0}^{\mathsf{T}} & \mathbf{C}\end{pmatrix}$$

[3 marks]

(b) Using the previous result compute $(\mathbf{UD})\mathbf{U}^{\mathsf{T}}$. Hence show $\boldsymbol{\Sigma} = \mathbf{UDU}^{\mathsf{T}}$.

$$(\mathbf{UD})\mathbf{U}^{\mathsf{T}} = \begin{pmatrix}\mathbf{A}-\mathbf{B}\mathbf{C}^{-1}\mathbf{B}^{\mathsf{T}} & \mathbf{B}\\ \mathbf{0}^{\mathsf{T}} & \mathbf{C}\end{pmatrix}\begin{pmatrix}\mathbf{I} & \mathbf{0}\\ \mathbf{C}^{-1}\mathbf{B}^{\mathsf{T}} & \mathbf{I}\end{pmatrix} = \begin{pmatrix}\mathbf{A} & \mathbf{B}\\ \mathbf{B}^{\mathsf{T}} & \mathbf{C}\end{pmatrix} = \boldsymbol{\Sigma}$$

[3 marks]

Answers

(c) Given that $\mathbf{\Sigma} = \mathbf{U}\mathbf{D}\mathbf{U}^{\mathsf{T}}$ write down $\mathbf{\Sigma}^{-1}$ in terms of $\mathbf{U}$ and $\mathbf{D}$

$$\mathbf{\Sigma}^{-1} = (\mathbf{U}^{\mathsf{T}})^{-1}\mathbf{D}^{-1}\mathbf{U}^{-1}$$

[1 mark]

(d) Demonstrate by direct multiplication that

$$\mathbf{U}^{-1} = \begin{pmatrix} \mathbf{I} & -\mathbf{B}\mathbf{C}^{-1} \\ \mathbf{0} & \mathbf{I} \end{pmatrix} \quad \mathbf{D}^{-1} = \begin{pmatrix} (\mathbf{A} - \mathbf{B}\mathbf{C}^{-1}\mathbf{B}^{\mathsf{T}})^{-1} & \mathbf{0} \\ \mathbf{0}^{\mathsf{T}} & \mathbf{C}^{-1} \end{pmatrix} \quad (\mathbf{U}^{\mathsf{T}})^{-1} = \begin{pmatrix} \mathbf{I} & \mathbf{0} \\ -\mathbf{C}^{-1}\mathbf{B}^{\mathsf{T}} & \mathbf{I} \end{pmatrix}$$

i.e. show $\mathbf{U}^{-1}\mathbf{U} = \mathbf{I}$, $\mathbf{D}^{-1}\mathbf{D} = \mathbf{I}$ and $(\mathbf{U}^{\mathsf{T}})^{-1}\mathbf{U}^{\mathsf{T}} = \mathbf{I}$.

(i)

$$\mathbf{U}^{-1}\mathbf{U} = \begin{pmatrix} \mathbf{I} & -\mathbf{B}\mathbf{C}^{-1} \\ \mathbf{0} & \mathbf{I} \end{pmatrix}\begin{pmatrix} \mathbf{I} & \mathbf{B}\mathbf{C}^{-1} \\ \mathbf{0} & \mathbf{I} \end{pmatrix} = \begin{pmatrix} \mathbf{I} & \mathbf{0} \\ \mathbf{0} & \mathbf{I} \end{pmatrix}$$

(ii)

$$\mathbf{D}^{-1}\mathbf{D} = \begin{pmatrix} (\mathbf{A} - \mathbf{B}\mathbf{C}^{-1}\mathbf{B}^{\mathsf{T}})^{-1} & \mathbf{0} \\ \mathbf{0}^{\mathsf{T}} & \mathbf{C}^{-1} \end{pmatrix}\begin{pmatrix} \mathbf{A} - \mathbf{B}\mathbf{C}^{-1}\mathbf{B}^{\mathsf{T}} & \mathbf{0} \\ \mathbf{0}^{\mathsf{T}} & \mathbf{C} \end{pmatrix} = \begin{pmatrix} \mathbf{I} & \mathbf{0} \\ \mathbf{0} & \mathbf{I} \end{pmatrix}$$

where we use $(\mathbf{A} - \mathbf{B}\mathbf{C}^{-1}\mathbf{B}^{\mathsf{T}})^{-1}(\mathbf{A} - \mathbf{B}\mathbf{C}^{-1}\mathbf{B}^{\mathsf{T}}) = \mathbf{I}$ and $\mathbf{C}^{-1}\mathbf{C} = \mathbf{I}$.

(iii)

$$(\mathbf{U}^{\mathsf{T}})^{-1}\mathbf{U}^{\mathsf{T}} = \begin{pmatrix} \mathbf{I} & \mathbf{0} \\ -\mathbf{C}^{-1}\mathbf{B}^{\mathsf{T}} & \mathbf{I} \end{pmatrix}\begin{pmatrix} \mathbf{I} & \mathbf{0} \\ \mathbf{C}^{-1}\mathbf{B}^{\mathsf{T}} & \mathbf{I} \end{pmatrix} = \begin{pmatrix} \mathbf{I} & \mathbf{0} \\ \mathbf{0} & \mathbf{I} \end{pmatrix}$$

[6 marks]

Answers

(e) Letting $\boldsymbol{z} = \binom{\boldsymbol{x}}{\boldsymbol{y}}$ then we can write

$$f_{\boldsymbol{X},\boldsymbol{Y}}(\boldsymbol{x},\boldsymbol{y}) = f_{\boldsymbol{Z}}(\boldsymbol{z}) = \mathcal{N}(\boldsymbol{z} \mid \boldsymbol{0},\boldsymbol{\Sigma}) = \frac{1}{\sqrt{\det(2\pi\boldsymbol{\Sigma})}} \mathrm{e}^{-\frac{1}{2}\boldsymbol{z}^{\mathsf{T}}\boldsymbol{\Sigma}^{-1}\boldsymbol{z}}.$$

where $\det(2\pi\boldsymbol{\Sigma})$ is the determinant of the matrix $2\pi\boldsymbol{\Sigma}$ and is introduced to ensure that $f_{\boldsymbol{X},\boldsymbol{Y}}(\boldsymbol{x},\boldsymbol{y})$ is normalised. From parts (c) and (d)

$$\boldsymbol{\Sigma}^{-1} = \begin{pmatrix} \mathbf{I} & \mathbf{0} \\ -\mathbf{C}^{-1}\mathbf{B}^{\mathsf{T}} & \mathbf{I} \end{pmatrix} \begin{pmatrix} (\mathbf{A} - \mathbf{B}\mathbf{C}^{-1}\mathbf{B}^{\mathsf{T}})^{-1} & \mathbf{0} \\ \mathbf{0}^{\mathsf{T}} & \mathbf{C}^{-1} \end{pmatrix} \begin{pmatrix} \mathbf{I} & -\mathbf{B}\mathbf{C}^{-1} \\ \mathbf{0} & \mathbf{I} \end{pmatrix}.$$

Expand out $\boldsymbol{z}^{\mathsf{T}}\boldsymbol{\Sigma}^{-1}\boldsymbol{z} = (\boldsymbol{x}^{\mathsf{T}},\boldsymbol{y}^{\mathsf{T}})\boldsymbol{\Sigma}^{-1}\binom{\boldsymbol{x}}{\boldsymbol{y}}$ (start by multiplying the vectors $\boldsymbol{z}^{\mathsf{T}}$ by $(\mathbf{U}^{\mathsf{T}})^{-1}$ and $\boldsymbol{z}$ by $\mathbf{U}^{-1}$)

$$\begin{aligned}
\boldsymbol{z}^{\mathsf{T}}\boldsymbol{\Sigma}^{-1}\boldsymbol{z} &= (\boldsymbol{x}^{\mathsf{T}},\boldsymbol{y}^{\mathsf{T}}) \begin{pmatrix} \mathbf{I} & \mathbf{0} \\ -\mathbf{C}^{-1}\mathbf{B}^{\mathsf{T}} & \mathbf{I} \end{pmatrix} \begin{pmatrix} (\mathbf{A} - \mathbf{B}\mathbf{C}^{-1}\mathbf{B}^{\mathsf{T}})^{-1} & \mathbf{0} \\ \mathbf{0}^{\mathsf{T}} & \mathbf{C}^{-1} \end{pmatrix} \begin{pmatrix} \mathbf{I} & -\mathbf{B}\mathbf{C}^{-1} \\ \mathbf{0} & \mathbf{I} \end{pmatrix} \begin{pmatrix} \boldsymbol{x} \\ \boldsymbol{y} \end{pmatrix} \\
&= (\boldsymbol{x}^{\mathsf{T}} - \boldsymbol{y}^{\mathsf{T}}\mathbf{C}^{-1}\mathbf{B}^{\mathsf{T}},\boldsymbol{y}^{\mathsf{T}}) \begin{pmatrix} (\mathbf{A} - \mathbf{B}\mathbf{C}^{-1}\mathbf{B}^{\mathsf{T}})^{-1} & \mathbf{0} \\ \mathbf{0}^{\mathsf{T}} & \mathbf{C}^{-1} \end{pmatrix} \begin{pmatrix} \boldsymbol{x} - \mathbf{B}\mathbf{C}^{-1}\boldsymbol{y} \\ \boldsymbol{y} \end{pmatrix} \\
&= (\boldsymbol{x}^{\mathsf{T}} - \boldsymbol{y}^{\mathsf{T}}\mathbf{C}^{-1}\mathbf{B}^{\mathsf{T}})(\mathbf{A} - \mathbf{B}\mathbf{C}^{-1}\mathbf{B}^{\mathsf{T}})^{-1}(\boldsymbol{x} - \mathbf{B}\mathbf{C}^{-1}\boldsymbol{y}) + \boldsymbol{y}^{\mathsf{T}}\mathbf{C}^{-1}\boldsymbol{y}
\end{aligned}$$

[4 marks]

Answers

(f) (We write $\int f(\boldsymbol{z})\mathrm{d}\boldsymbol{z}$ for $\int\cdots\int f(\boldsymbol{z})\mathrm{d}z_1\cdots\mathrm{d}z_n$.) To compute $f_{\boldsymbol{X}|\boldsymbol{Y}}(\boldsymbol{x} \mid \boldsymbol{y}) = f_{\boldsymbol{X},\boldsymbol{Y}}(\boldsymbol{x},\boldsymbol{y})/f_{\boldsymbol{Y}}(\boldsymbol{y})$ we need to find

$$\begin{aligned} f_{\boldsymbol{Y}}(\boldsymbol{y}) &= \int_{-\infty}^{\infty} f(\boldsymbol{x},\boldsymbol{y})\mathrm{d}\boldsymbol{x} \\ &= \int_{-\infty}^{\infty} \frac{1}{\sqrt{\det(2\pi\boldsymbol{\Sigma})}} \mathrm{e}^{-\frac{1}{2}(\boldsymbol{x}^{\mathsf{T}}-\boldsymbol{y}^{\mathsf{T}}\mathbf{C}^{-1}\mathbf{B}^{\mathsf{T}})(\mathbf{A}-\mathbf{B}\mathbf{C}^{-1}\mathbf{B}^{\mathsf{T}})^{-1}(\boldsymbol{x}-\mathbf{B}\mathbf{C}^{-1}\boldsymbol{y})-\frac{1}{2}\boldsymbol{y}^{\mathsf{T}}\mathbf{C}^{-1}\boldsymbol{y}}\mathrm{d}\boldsymbol{x} \\ &= \frac{\mathrm{e}^{-\frac{1}{2}\boldsymbol{y}^{\mathsf{T}}\mathbf{C}^{-1}\boldsymbol{y}}}{\sqrt{\det(2\pi\boldsymbol{\Sigma})}} \int_{-\infty}^{\infty} \mathrm{e}^{-\frac{1}{2}(\boldsymbol{x}^{\mathsf{T}}-\boldsymbol{y}^{\mathsf{T}}\mathbf{C}^{-1}\mathbf{B}^{\mathsf{T}})(\mathbf{A}-\mathbf{B}\mathbf{C}^{-1}\mathbf{B}^{\mathsf{T}})^{-1}(\boldsymbol{x}-\mathbf{B}\mathbf{C}^{-1}\boldsymbol{y})}\mathrm{d}\boldsymbol{x} \end{aligned}$$

By making a change of variable from $\boldsymbol{x}$ to $\boldsymbol{u} = \boldsymbol{x} - \mathbf{B}\mathbf{C}^{-1}\boldsymbol{y}$ rewrite the integral

$$I = \int_{-\infty}^{\infty} \mathrm{e}^{-\frac{1}{2}(\boldsymbol{x}^{\mathsf{T}}-\boldsymbol{y}^{\mathsf{T}}\mathbf{C}^{-1}\mathbf{B}^{\mathsf{T}})(\mathbf{A}-\mathbf{B}\mathbf{C}^{-1}\mathbf{B}^{\mathsf{T}})^{-1}(\boldsymbol{x}-\mathbf{B}\mathbf{C}^{-1}\boldsymbol{y})}\mathrm{d}\boldsymbol{x}$$

then use

$$\int_{-\infty}^{\infty} \mathrm{e}^{-\frac{1}{2}\boldsymbol{u}^{\mathsf{T}}\mathbf{M}^{-1}\boldsymbol{u}}\mathrm{d}\boldsymbol{u} = \sqrt{\det(2\pi\mathbf{M})}$$

to evaluate $f_{\boldsymbol{Y}}(\boldsymbol{y})$.

Making the change of variables $\boldsymbol{u} = \boldsymbol{x} - \mathbf{B}\mathbf{C}^{-1}\boldsymbol{y}$

$$I = \int_{-\infty}^{\infty} \mathrm{e}^{-\frac{1}{2}\boldsymbol{u}^{\mathsf{T}}(\mathbf{A}-\mathbf{B}\mathbf{C}^{-1}\mathbf{B}^{\mathsf{T}})^{-1}\boldsymbol{u}}\mathrm{d}\boldsymbol{u} = \sqrt{\det(2\pi(\mathbf{A}-\mathbf{B}\mathbf{C}^{-1}\mathbf{B}^{\mathsf{T}}))}$$

Thus

$$f_{\boldsymbol{Y}}(\boldsymbol{y}) = \sqrt{\frac{\det(2\pi(\mathbf{A}-\mathbf{B}\mathbf{C}^{-1}\mathbf{B}^{\mathsf{T}}))}{\det(2\pi\boldsymbol{\Sigma})}}\mathrm{e}^{-\frac{1}{2}\boldsymbol{y}^{\mathsf{T}}\mathbf{C}^{-1}\boldsymbol{y}}.$$

where $\det(\boldsymbol{\Sigma})$ denotes the determinant of a matrix $\boldsymbol{\Sigma}$. In passing we recall that $\boldsymbol{\Sigma} = \mathbf{U}\mathbf{D}\mathbf{U}^{\mathsf{T}}$ so $\det(\boldsymbol{\Sigma}) = \det(\mathbf{U})\det(\mathbf{D})\det(\mathbf{U}^{\mathsf{T}})$, But it is easy to show $\det(\mathbf{U}^{\mathsf{T}}) = \det(\mathbf{U}) = 1$ and because $\mathbf{D}$ is block diagonal $\det(\mathbf{D}) = \det(\mathbf{A}-\mathbf{B}\mathbf{C}^{-1}\mathbf{B}^{\mathsf{T}})\det(\mathbf{C})$ so that $\det(\boldsymbol{\Sigma}) = \det(\mathbf{A}-\mathbf{B}\mathbf{C}^{-1}\mathbf{B}^{\mathsf{T}})\det(\mathbf{C})$. Thus

$$f_{\boldsymbol{Y}}(\boldsymbol{y}) = \frac{1}{\sqrt{\det(2\pi\mathbf{C})}}\mathrm{e}^{-\frac{1}{2}\boldsymbol{y}^{\mathsf{T}}\mathbf{C}^{-1}\boldsymbol{y}} = \mathcal{N}(\boldsymbol{y} \mid \mathbf{0},\mathbf{C}).$$

Recall that

$$f_{\boldsymbol{X},\boldsymbol{Y}}(\boldsymbol{x},\boldsymbol{y}) = \mathcal{N}\left(\begin{pmatrix}\boldsymbol{x}\\ \boldsymbol{y}\end{pmatrix}\middle|\begin{pmatrix}\mathbf{0}\\ \mathbf{0}\end{pmatrix},\begin{pmatrix}\mathbf{A} & \mathbf{B}\\ \mathbf{B}^{\mathsf{T}} & \mathbf{C}\end{pmatrix}\right)$$

so we see $\boldsymbol{y}$ is normally distributed with covariance $\mathbf{C}$ as might be expected.

[5 marks]

Answers

(g) Using $f_{\boldsymbol{X}|\boldsymbol{Y}}(\boldsymbol{x} \mid \boldsymbol{y}) = f_{\boldsymbol{X},\boldsymbol{Y}}(\boldsymbol{x},\boldsymbol{y})/f_{\boldsymbol{Y}}(\boldsymbol{y})$ write down $f_{\boldsymbol{X}|\boldsymbol{Y}}(\boldsymbol{x} \mid \boldsymbol{y})$.

$$
\begin{aligned}
f_{\boldsymbol{X}|\boldsymbol{Y}}(\boldsymbol{x} \mid \boldsymbol{y}) &= \frac{f_{\boldsymbol{X},\boldsymbol{Y}}(\boldsymbol{x},\boldsymbol{y})}{f_{\boldsymbol{Y}}(\boldsymbol{y})} = \frac{\det(2\pi\mathbf{C})^{\frac{1}{2}}}{\det(2\pi\boldsymbol{\Sigma})^{\frac{1}{2}}} \mathrm{e}^{-\frac{1}{2}(\boldsymbol{x}^\mathsf{T}-\boldsymbol{y}^\mathsf{T}\mathbf{C}^{-1}\mathbf{B}^\mathsf{T})(\mathbf{A}-\mathbf{B}\mathbf{C}^{-1}\mathbf{B}^\mathsf{T})^{-1}(\boldsymbol{x}-\mathbf{B}\mathbf{C}^{-1}\boldsymbol{y})} \\
&= \frac{1}{\det(2\pi(\mathbf{A}-\mathbf{B}\mathbf{C}^{-1}\mathbf{B}^\mathsf{T}))^{\frac{1}{2}}} \mathrm{e}^{-\frac{1}{2}(\boldsymbol{x}^\mathsf{T}-\boldsymbol{y}^\mathsf{T}\mathbf{C}^{-1}\mathbf{B}^\mathsf{T})(\mathbf{A}-\mathbf{B}\mathbf{C}^{-1}\mathbf{B}^\mathsf{T})^{-1}(\boldsymbol{x}-\mathbf{B}\mathbf{C}^{-1}\boldsymbol{y})} \\
&= \mathcal{N}(\boldsymbol{x} \mid \mathbf{B}\mathbf{C}^{-1}\boldsymbol{y}, \mathbf{A}-\mathbf{B}\mathbf{C}^{-1}\mathbf{B}^\mathsf{T}).
\end{aligned}
$$

This is the conditioning result we use in deriving the Bayesian update for a Gaussian Process.

[3 marks]

End of question 4

END OF PAPER

Answers